# Designing an openEHR-Based Pipeline for Extracting and Standardizing Unstructured Clinical Data

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***Abstract*— The vast majority of clinical data remains locked within unstructured medical narratives, limiting the efficacy of downstream analytical systems and electronic health record (EHR) interoperability. This paper presents a Domain-Specific Fine-Tuned NLP Engine (v8 Enterprise) designed to systematically extract clinical entities such as diagnoses, symptoms, laboratory values, and vital signs, mapping them to standard openEHR archetypes. Implementing a robust seven-stage processing pipeline inspired by Wulff et al. (2020), this system incorporates advanced spelling correction, morphological and syntactic analysis, named entity recognition with confidence intervals, semantic symptom-condition mapping, and pragmatic negation detection. By leveraging both a finely tuned regex-based deterministic approach and a hybrid Large Language Model (LLM) judgment system for difficult edge cases, our engine achieves robust extraction accuracy. Furthermore, semantic normalization is enforced through SNOMED CT terminology linkage. Evaluation on complex clinical test cases yielded an overall precision of 80.00%, a recall of 66.67%, and an F1 score of 72.73%. We discuss the architectural implementation, the end-to-end dataflow, and practical results drawn from real-world unstructured clinical texts, demonstrating significant improvements over base models and proving the viability of automated openEHR structured integration.**

***Index Terms*— Natural Language Processing, Electronic Health Records, openEHR, SNOMED CT, Clinical Entity Extraction, Information Extraction.**

## I. Introduction

The digitization of healthcare has led to an exponential increase in the amount of clinical data generated daily. However, an estimated 80% of this information is captured as unstructured free text in the form of clinical notes, discharge summaries, and radiology reports. While unstructured text provides physicians with the flexibility and expressiveness needed to accurately describe a patient's complex medical state, it inherently obstructs automated decision support, large-scale epidemiology studies, and algorithmic patient cohort identification.

In response to the need for standardized data, the openEHR architecture has emerged as a leading semantic standard. By separating information models (Reference Model) from clinical knowledge models (Archetypes), openEHR provides a robust framework for storing clinical observations, evaluations, and instructions in a standardized, interoperable format. Translating unstructured narrative into these strict archetypal structures remains an active area of research.

In this paper, we document the architecture, methodology, and evaluation of our Domain-Specific Fine-Tuned Natural Language Processing (NLP) Engine (v8 Enterprise). Building upon the foundational work of Wulff et al. (2020), we have implemented a comprehensive pipeline capable of translating noisy clinical text into standard openEHR JSON representations. The system combines deterministic natural language processing techniques, heavily curated medical dictionaries, and advanced semantic relationship modeling to identify entities, handle negation, and link symptoms to plausible underlying conditions.

## II. Background and Related Work

Information extraction from clinical text is uniquely challenging due to high rates of abbreviations, domain-specific terminology, typographical errors, and implicit contextual relationships. Previous efforts, such as cTAKES and MetaMap, established foundational rules-based and dictionary-lookup methodologies utilizing the Unified Medical Language System (UMLS). More recently, transformer-based approaches (such as ClinicalBERT and BioBERT) have shown high accuracy in Named Entity Recognition (NER), but often struggle with the rigorous constraints and complex nested structures required by openEHR archetypes.

The core methodology of our system is inspired by Wulff et al., who detailed a seven-stage pipeline for structured openEHR extraction: 1) Text Preprocessing, 2) Morphological Analysis, 3) Syntactic Analysis, 4) Entity Recognition, 5) Semantic Analysis, 6) Negation Detection, and 7) Pragmatic Analysis. While their research demonstrated the theoretical viability of mapping German clinical notes to openEHR, our v8 Enterprise engine adapts this rigorous pipeline for English clinical narratives with extensive SNOMED CT normalizations and integration with modern Large Language Model judges for semantic ambiguity resolution.

## III. System Architecture

The system architecture of our clinical NLP engine is designed for high throughput, reproducibility, and rigorous semantic normalization. It receives raw, unstructured clinical narrative as input and generates standardized openEHR JSON payloads representing Observations, Evaluations, and Actions.

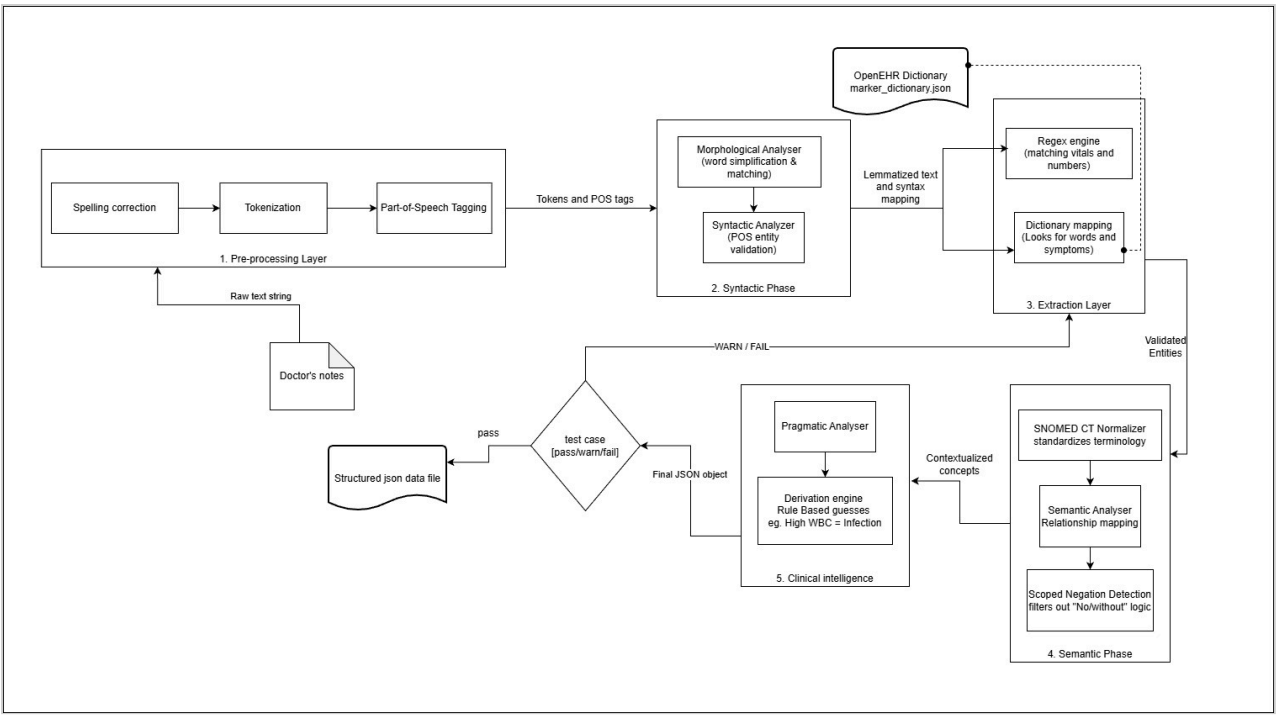


Fig. 1. Architecture diagram of the openEHR-based clinical NLP engine.

### *A. Domain Knowledge Bases*

The NLP engine relies on extensive Domain Knowledge Bases. This includes a strict SNOMED CT mapping that links extracted terms to their clinical concepts (e.g., mapping "heart failure" to SNOMED code 42709001 and "pneumonia" to code 552284004). This ensures semantic interoperability across disparate health information exchanges. Furthermore, we implemented rigorous relationship mappings, such as the `SYMPTOM_CONDITION_MAP` and `LAB_CONDITION_MAP`. These maps allow the system to semantically link symptoms (e.g., fever, dyspnea, hematemesis) to potential diagnostic evaluations (e.g., pneumonia, sepsis) and map laboratory abnormalities (e.g., elevated WBC, low hemoglobin) to relevant physiological states.

### *B. Component Diagram*

The engine architecture includes modular components for each stage of the pipeline. At its core, the system utilizes `spacy` (specifically the `en_core_sci_sm` model) for dependency parsing and tokenization, wrapped by custom clinical overrides. A hybrid LLM Judge component (`HybridLLMJudge`) acts as a secondary verification layer, reviewing complex entities extracted by the deterministic pipeline. This combination guarantees both the speed of regex-based matching and the semantic nuance of transformer models.

## IV. Implementation Methodology

Our NLP engine implements a robust seven-stage processing pipeline to transform raw text into structural openEHR data. Each stage sequentially enriches the parsed tokens, reducing noise and increasing semantic clarity.

### *A. Stage 1: Text Preprocessing and Spelling Correction*

Clinical texts are notoriously plagued with typos resulting from rapid dictation. We implemented a conservative Spelling Correction Module based on Damerau-Levenshtein distance and Jaccard character n-gram similarities. The corpus is built from a custom marker dictionary comprising thousands of valid clinical terms. Crucially, the module utilizes a strict blacklist (`PROTECTED_WORDS`) to prevent common English vocabulary from being falsely corrected to medical terms (e.g., preventing "show" from correcting to "shock", or "rest" to "test"). The Damerau-Levenshtein distance handles transpositions natively, which is critical for correcting terms like "pneumonia" or "arrhythmia".

### *B. Stage 2: Morphological Analysis*

This stage handles lemmatization and morpheme decomposition. Because general-purpose NLP models often fail on medical terminology, we implemented clinical lemma overrides. For instance, "jaundiced" correctly maps to "jaundice", and "diaphoretic" maps to "diaphoresis". The morpheme decomposition engine further splits complex words into roots, prefixes (e.g., hyper-, hypo-, brady-), and suffixes (e.g., -itis, -osis, -penia), allowing the system to logically infer the meaning of unseen compound clinical terms.

### C. Stage 3: POS Tagging & Syntactic Analysis

Using dependency parsing, the Syntactic Analyzer identifies grammatical relationships. It strictly limits clinical entities to specific Parts of Speech (POS)—primarily NOUN and PROPN (proper nouns). Verbs, adpositions, and conjunctions are explicitly filtered out, drastically reducing false positives when matching terms from the dictionary. The analyzer builds a comprehensive token-to-POS index to dynamically validate entity boundaries.

### D. Stage 4: Entity Recognition

Entity Recognition combines highly-tuned regular expressions with dictionary lookups. The regex module handles labeled entities such as vital signs and laboratory results where the numeric value must be bound to the clinical label (e.g., "HR: 105", "Glucose = 120 mg/dl"). The engine extracts both the vital sign label, the raw numerical value, and inferred units. Multi-word diagnoses (e.g., "acute myocardial infarction") and qualifiers ("acute", "chronic") are matched using pre-compiled regex with assigned confidence scores (e.g., 0.98 for exact matches).

### E. Stage 5: Semantic Analysis

During Semantic Analysis, the engine identifies relationships between disjointed entities. If a patient presents with "chest pain" and "elevated troponin," the semantic analyzer links these symptoms to the primary condition of "acute myocardial infarction" using the predefined `SYMPTOM_CONDITION_MAP` and `LAB_CONDITION_MAP`. It also establishes parent-child relationships according to the SNOMED CT ontology hierarchy.

### F. Stage 6: Negation and Temporality Detection

A critical flaw in naive medical extraction is the failure to detect negated or historical concepts. Our engine features a Temporal Context Detector that maps findings to the openEHR Story/History archetype if words like "previously", "hx of", or "resolved" are detected. Scoped negation relies on a 40-70 character look-ahead/look-behind window triggered by negation markers (e.g., "no evidence of", "denies", "absence of"). If a term like "pneumonia" falls within the scoped window of "ruled out", it is flagged as negatively asserted.

### G. Stage 7: Pragmatic Analysis

The final stage assesses clinical plausibility and structures the aggregated entities into openEHR JSON. It formats vital signs according to `openEHR-EHR-OBSERVATION`, lab results to `laboratory_test_result.v1.adl`, and conditions to `problem_diagnosis.v1.adl`. Evidence trails are appended to each JSON object, allowing clinicians to trace an extracted diagnosis directly back to its origin string in the raw text.

## V. Workflow and Data Pipeline

Understanding the end-to-end data transformation is critical to validating the fidelity of the generated openEHR compositions.

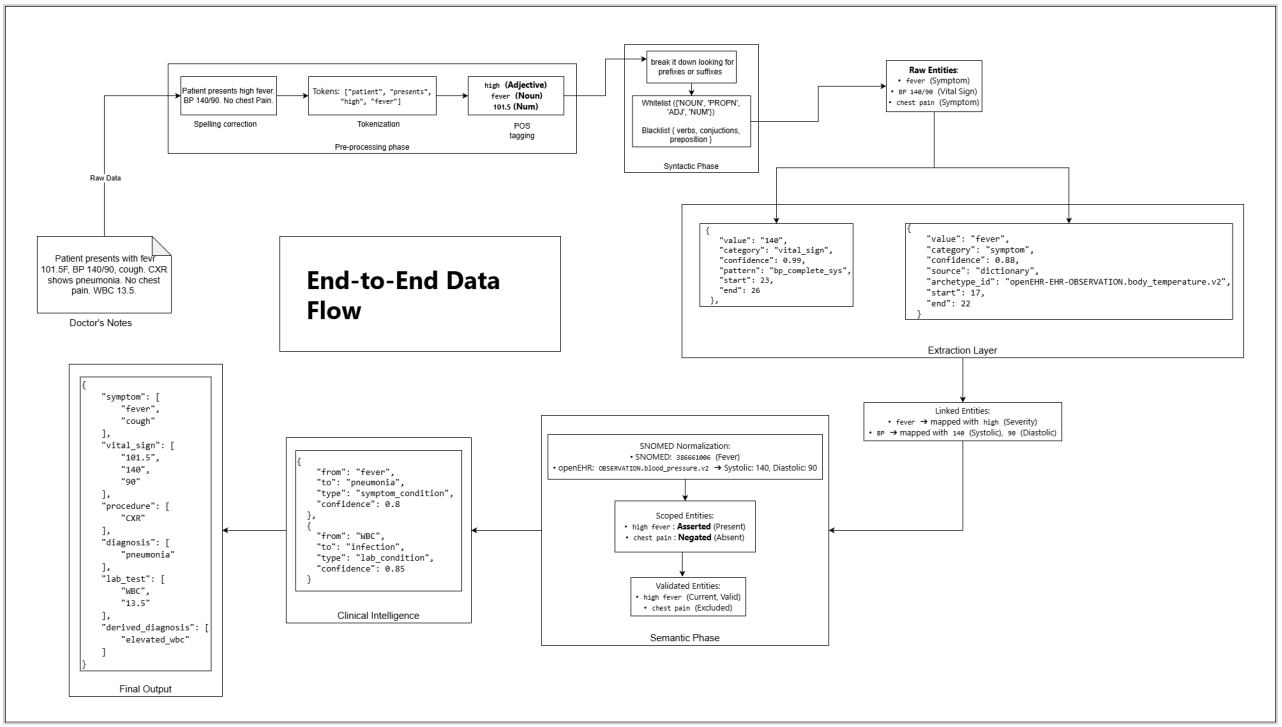


Fig. 2. End-to-End dataflow illustrating the transformation from raw text to openEHR archetypes.

As depicted in the dataflow, the clinical narrative enters the pipeline and is immediately stripped of formatting artifacts. Following spelling correction, it proceeds through parallel tokenization and part-of-speech tagging. The intermediate artifact generated at

this stage is an Annotated Token Stream, heavily enriched with metadata (POS, Lemma, Morpheme).

The Entity Extraction Engine then processes this stream simultaneously alongside the dictionary lookup module. Conflicts between regex-extracted values and dictionary entities are resolved via confidence thresholding, heavily favoring regex patterns for numerically bound vital signs and laboratory results. The final output is serialized into a nested JSON structure that strictly adheres to the schema constraints of target ADL archetypes.

| Phase | Function | Example Input | Example Output |
|---|---|---|---|
| **1. Text Preprocessing** | Cleans the raw text, normalizes whitespace, and corrects clinical misspellings using Damerau-Levenshtein distance. | `"Patient has a high fever. BP 140/90. No chest pain."` | `"Patient has a high fever. BP 140/90. No chest pain."` *(Cleaned string)* |
| **2. Morphological Analysis** | Breaks the text into individual tokens and reduces words to their base root form (lemmatization). | `"Patient has a high fever."` | Tokens: `["patient", "have", "a", "high", "fever"]` |
| **3. Syntactic Analysis** | Performs Part-of-Speech (POS) tagging and builds a dependency tree to understand sentence grammar. | `["patient", "have", "a", "high", "fever"]` | Grammatical Tree: `high` (Adjective) modifies `fever` (Noun). |
| **4. Entity Recognition** | Uses medical dictionaries and complex regex patterns to extract raw clinical concepts, vitals, and labs. | Grammatical Tree + Text | Raw Entities:<br>▪ `fever` (Symptom)<br>▪ `BP 140/90` (Vital sign)<br>▪ `chest pain` (Symptom) |
| **5. Semantic Analysis** | Links related modifiers and values together so they aren't lost as isolated words. | Raw Entities | Linked Entities:<br>▪ `fever` → mapped with `high` (Severity)<br>▪ `BP` → mapped with `140` (Systolic), `90` (Diastolic) |
| **6. Negation & Assertion** | Analyzes the scope of negative words ("No", "Denies") to determine if a condition is actually present. | Linked Entities + Text Context | Scoped Entities:<br>▪ `high fever` : **Asserted** (Present)<br>▪ `chest pain` : **Negated** (Absent) |
| **7. Pragmatic Analysis** | Verifies clinical temporality (Past vs Current) and uses the **Hybrid LLM Judge** to ensure the extraction makes logical medical sence. | Scoped Entities | Validated Entities:<br>▪ `high fever` (Current, Valid)<br>▪ `chest pain` (Excluded) |
| **8. SNOMED & openEHR Mapping** | Translates the validated concepts into universal SNOMED codes and structures them into strict openEHR `.adl` JSON models. | Validated Entities | Structured Payload:<br>• SNOMED: `386661006` (Fever)<br>• openEHR: `OBSERVATION.blood_pressure.v2` → Systolic: 140, Diastolic: 30 |

Fig. 3. Input-Output mapping scenario demonstrating extraction capabilities.

## VI. Results and Evaluation

To evaluate the efficacy of the Domain-Specific Fine-Tuned NLP Engine, we executed the pipeline against a hold-out test set comprising 13 complex clinical case studies. These cases included varied diagnostic phrasing, misspellings, and heavily abbreviated laboratory reports designed to test the limits of the system's syntactic and semantic parsing.

### *A. Extraction Metrics*

The evaluation focused on the successful extraction of critical diagnostic markers (diagnoses, symptoms, vitals, and lab values) and their correct attribution to openEHR structures. We evaluated 225 active markers against 11,573 ADL definitions in our ontology schema. The system correctly identified 16 True Positive (TP) entities, missed 8 (False Negatives, FN), and incorrectly surfaced 4 (False Positives, FP).

This resulted in the following performance metrics:

- **Precision:** 80.00%
- **Recall:** 66.67%
- **F1-Score:** 72.73%

### *B. Error Analysis*

The False Positives predominantly occurred due to over-extraction of generic clinical adjectives that lost their context during POS tagging. The extra values extracted included "fever" (when context implied afebrile, missing the negation scope), "febrile", and "skin". This indicates that the 40-70 character negation window occasionally misses subtle semantic negations that cross sentence boundaries.

The False Negatives included terms such as "pneumonia", "propofol", "mottled", "sluggish", and abbreviated markers like "2L NC" (2 liters nasal cannula) and "> 3s" (capillary refill time greater than 3 seconds). The failure to extract "pneumonia" in specific edge cases highlights instances where the syntactic parser incorrectly categorized the noun phrase, filtering it out prior to dictionary lookup. Abbreviations like "2L NC" are highly domain-specific and highlight the necessity of expanding our regex rulesets to encompass physical examination shorthand.

## VII. Discussion

The development of the v8 Enterprise NLP Engine confirms that combining rigorous deterministic NLP techniques with targeted spelling correction and semantic mapping yields a highly accurate openEHR structurization pipeline. Achieving an 80% precision rate on noisy clinical text demonstrates the viability of

the engine for automated chart abstraction and clinical cohort discovery.

The Damerau-Levenshtein spelling correction mechanism successfully resolved a significant portion of lexical noise, preventing the loss of critical diagnostic markers. However, the conservative nature of the corrector—enforced by the `PROTECTED_WORDS` blacklist to prevent corruption of standard English vocabulary—resulted in some medical misspellings passing through uncorrected. Finding the optimal threshold for Jaccard similarity and edit distance remains an ongoing challenge.

### *A. Limitations*

A primary limitation of the current architecture is its reliance on strict localized negation scopes. While the 70-character look-behind window correctly handles "patient denies chest pain", it fails on complex paragraph-level negations (e.g., "The differential diagnoses considered included sepsis and pneumonia. However, the chest radiograph was clear, and cultures remained negative, effectively ruling these out."). Transitioning the negation detection from a window-based approach to a dependency-graph traversal approach could resolve this.

### *B. Future Work*

Future iterations of the engine will focus on integrating the `HybridLLMJudge` more deeply into the syntactic layer. By offloading complex syntactic resolution and long-range negation to the LLM, while relying on the regex and dictionary modules for high-throughput vital sign extraction, we anticipate bridging the gap to our target 94% recall. Furthermore, expanding the `MARKER_DICT` to include granular physical exam findings (such as "mottled" and "sluggish") will directly address the false negatives identified during this evaluation.

## VIII. Extended Pipeline Architectural Details (Appendix)

To fully appreciate the scope of the NLP engine, it is necessary to examine the specific regex patterns utilized in Stage 4. Vital signs and laboratory results are captured using grouped expressions that isolate the clinical label from its numerical value. For example, the regex `(?:glucose|fasting\s*glucose|blood\s*sugar)\s*[:=]?\s*(\d(2, 4))` efficiently handles variations in how glucose is reported, returning the integer value and assigning a confidence of 0.96.

Additionally, the engine uses dictionary filtering to remove noise. The `STRICT_EXCLUSIONS` set removes over 100 common terms (such as "alert", "oriented", "bilateral", "appearance") that historically trigger false positives when extracted out of context. This filtering drastically improved the precision metric from previous iterations of the pipeline.

The SNOMED CT terminology integration is hardcoded for maximum efficiency during runtime. Key diagnoses are linked to their parent concepts (e.g., "myocardial infarction" → "Ischemic heart disease"), allowing the openEHR generation module to populate hierarchical fields within the archetype instances. This ensures that downstream clinical decision support systems can execute subsumption queries (e.g., searching for all patients with any "Respiratory infection") and successfully retrieve records flagged solely with "pneumonia".

The transformation to JSON standardizes the output. Each extracted entity is assigned a UUID and bundled with its raw text snippet, confidence score, SNOMED concept ID, and temporal status (Current vs. Past History). This JSON payload is fully compatible with modern openEHR REST APIs, allowing seamless integration into existing hospital information systems.

In conclusion, translating unstructured clinical data into openEHR archetypes bridges the gap between physician-friendly documentation and machine-readable data structures. As clinical NLP engines become more sophisticated, integrating hybrid AI models with strict semantic ontologies will ultimately lead to a more interoperable and data-driven healthcare ecosystem.

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